

Rhea Pandita

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EXPERIENCE

USC CeNEC A-Z Lab | Graduate Research Assistant

Aug 2025–Present | Los Angeles, CA

- Built a rate-limited LLM inference pipeline (deque-based scheduler + retry/backoff) enabling reliable batch evaluation across **GPT-OSS-120B**, **Claude-Opus-4.x**, **LLaMA-3.3-70B**, and **Qwen-32B**.
- Architected a modular evaluation framework (prompt orchestration, API parsing, structured logging), achieving **98% output extraction** across 158 annotated multimodal samples.
- Developed automated logging system (Python, Pandas) tracking per-sample outputs and aggregate metrics for reproducible cross-model benchmarking at scale.

KPIT Technologies | Machine Learning Intern

Jun 2024–Aug 2024 | Bangalore, India

- Engineered a Python-based **CAD processing pipeline** to parse IGES/STEP files and extract structured geometry for automated backend modification of 3D designs.
- Built geometric reconstruction algorithms mapping 2D shapes to 3D counterparts, enabling scalable CAD parameter editing across multiple files.

KPIT Technologies | Machine Learning Intern

May 2023–Aug 2023 | Bangalore, India

- Built a text-to-image generation system using **Stable Diffusion v1-4** (HuggingFace Diffusers), leveraging **CLIP text encoder**, **latent U-Net**, and **VAE** to generate automotive design concepts from natural language prompts.
- Developed an **image-to-3D pipeline** using **GLPN (vinvino02/glpn-nyu)** (monocular depth estimation) and **Open3D** (Poisson reconstruction) to output **.obj meshes** for rapid prototyping.

SKILLS

Languages: Python, SQL, C, JavaScript, TypeScript

AI/ML: LLMs, RAG, Transformers, HuggingFace, LangChain, FAISS, RAGAS, ONNX

Libraries: PyTorch, Pandas, NumPy, Scikit-learn, Matplotlib, NLTK, SpaCy

Frameworks/Tools: FastAPI, React, Node.js, Flask, Docker, Git

Infrastructure: AWS EC2, Redis, Kafka, Streamlit Cloud

Databases: MySQL, SQLite

Courses: Deep Learning, Reinforcement Learning, NLP, Machine Learning, Statistical Inference, Data Mining, Big Data Analysis, Database Systems, Information Retrieval & Web Search Engines

PROJECTS

Streamora: AI-Powered Streaming Recommendation System 🧠🔗 | FastAPI, Redis, Kafka, Docker, AWS EC2

- Trained **BERT4Rec** and **SASRec** on **110K** user sequences across **9,387** items with a **38-dim** user feature vector for cold-start personalisation; exported to **ONNX**.
- Deployed full-stack recommendation platform on **AWS EC2** via **FastAPI** with **Redis** caching, achieving **2ms** hit vs **150ms** inference, and **Kafka** async feedback pipeline decoupled from the request-response cycle.
- Implemented hybrid semantic search (**384-dim MiniLM** embeddings+ **TMDB API**) with a **React** frontend featuring per-profile data isolation, WebSocket watch party, and real-time feedback loop updating model inputs.

AI-Assisted Personal Finance Management System 🧠🔗 | Python, LangChain, FAISS, Groq API, MySQL, Streamlit

- Deployed a cloud-hosted RAG chatbot system on Streamlit Cloud by migrating from ChromaDB to FAISS with **LLaMA-3.3-70B**, reducing response latency **98%** (60s → 0.57s) across **4,530 transactions**.
- Improved RAG evaluation scores (faithfulness **0.81**→**0.95**, semantic similarity **0.61**→**0.73**) by replacing Mistral+ChromaDB with LLaMA-3.3-70B+FAISS over a curated **379-entry** knowledge base.
- Curated a **330 question–SQL** dataset for prompt refinement; resolved cross-platform deployment blockers enabling the previously local-only application to run on Linux cloud infrastructure.

ICU Deterioration Prediction 🧠 | Python, PyTorch, ClinicalBERT, BioViL-T, Scikit-learn, Pandas, SHAP

- Prevented **0.15–0.20 AUROC** inflation by designing a **feature-cutoff** leakage barrier across **91k** stays and **35m** clinical records
- Achieved **AUROC 0.9998 (89% recall)** on respiratory failure) by fine-tuning **ClinicalBERT** on serialized patient notes with strict patient-level splits and val-guided epoch selection.
- Ran a **7-condition ablation** across modality combinations; identified tabular+chest X-ray features as optimal (**AUROC 0.8700**) by fusing structured data with **PCA-compressed BioViL-T** image embeddings.
- Built a **Streamlit** clinical decision support UI with **SHAP** feature attribution and colour-coded risk gauge for interpretable bedside risk assessment.

EDUCATION

University of Southern California

Master of Science in Computer Science

Los Angeles, CA

Aug 2025–May 2027

Shiv Nadar University

B.Tech in Artificial Intelligence & Data Science

Chennai, India

Sep 2021–Jul 2025

PUBLICATION

- T.R. Althi, V. Kothamachu, **R. Pandita**, "Accelerating Automotive Design: Harnessing AI Models for Efficient 3D Design and Development of Automobile Systems and Subsystems" *IEEE ITEC-India*, 2023. doi:10.1109/ITEC-India59098.2023.10471378